Predicting Hate Speech Scores in Bluesky Posts

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Project Overview and Goals

The rise of social media platforms over the past 20 years has fundamentally changed the way humans communicate. Platforms such as Facebook have seen exponential growth, increasing from 1.4 billion monthly active users in 2014 to over 3 billion in 2024 (Kumar, 2025). Simultaneously, new platforms like Bluesky (2024), Mastodon (2016) and Meta's Threads (2023) have surged, each offering different levels and methods of decentralization, moderation or control. As the number of social media platforms increases, so does the volume of online content. Consequentially, this increase in online content has also brought with it a stark growth of harmful, hateful and abusive language (Immenkamp et al., 2024).

Recent studies, like *Online hate speech victimization [...]* by Dreißigacker have shown, that the exposure to hate speech online dramatically increases the probability to develop mental health problems, like anxiety, depression, and increased suicide risk among targeted individuals (Dreißigacker et al., 2024). Social media platforms now face growing public and legal pressure to better manage this growing issue. The necessity for social media platforms to protect its users from harmful content and the need to comply with stricter international regulations, like the European Digital Services Acs (DSA) and Australia's Online Safety Act, have led companies to constantly improve their content moderation services.

From a societal perspective, our project contributes to online safety efforts by attempting to create a tool to help detect hateful speech in recent comments. This, in turn, could provide an additional safety feature to ensure that such hate comments reach a smaller audience, thus aiding in maintaining the safety of users navigating online.

This project can also bring value from a business perspective, as a well moderated platform helps maintain public image, user-trust and avoid legal disputes. Furthermore, by using machine learning to analyze comments and flagging those that have a high likelihood to contain hate speech, business could cut down on the number of moderators necessary to review individual cases and likely reducing moderation cost (Objectways. 2024).

Our project qualifies as a big data project, since it falls into the fundamental criteria of Big Data by fulfilling the aspects of **Volume, Velocity, Variety** and **Value**:

- Volume: The training dataset used throughout this project consists of 131 variables and over 135,000 rows of labelled comments, aggregated from across three different platforms (Kennedy et al., 2020). Additionally, the project accesses real-world data with help of the Bluesky API, which has the capacity to retrieve thousands of recent user-generated comments, further increasing the project's scale
- Velocity: Our system is designed with the assumption that thousands of new comments could arrive in a very short time span. While our current implementation is based on batches of comments (meaning that each time the program executes, a specified number of new comments are fetched), it uses a Spark environment to ensure scalable processing, which could be adapted to process a continuous stream of content.
- Variety: The project makes use of two different data sources. The data used to train the model is a structured dataset of labelled comments, presented in a CSV file. On the other hand, the data fetched from Bluesky represents a semi-structured dataset of raw posts. The contents of this dataset are presented in a JSON structure and contain metadata that needs to be cleaned. This combination creates a variety in both data format and source of data.

- Value: The project offers practical value in a very practical way. Hate speech is on the rise on Bluesky's main competitor, X, and we see a reliable detector of it as important to combat this. We think that hate speech cannot exist under the guise of free speech and view safety as a cornerstone of social platforms, such as Bluesky (Manke, 2025).

The objective of this project is to develop a scalable machine learning pipeline for detecting hate speech in comments made on Bluesky using a regression-based approach. The primary goal is to train three different regression models, capable of predicting a continuous hate speech score of Bluesky comments. This involves obtaining correct data for training and testing, applying suitable preprocessing methods, and constructing prediction models using Apache Spark. Real world applicability will be tested by using live, semi-structured data retrieved from Bluesky through its API. Additionally, this project aims to address the legal and ethical challenges that might arise with the utilization of live data.

Data Sources & Architecture

To achieve the development of our predictive modeling, we used two distinct data sources.

Dataset 1: Labeled Hate Speech Corpus

To train our predictive model, we made use of a publicly available dataset compiled by researchers at the University of California, Berkeley (Kennedy et al., 2020). This dataset contains over 135,000 labeled entries, created from comments sourced from YouTube, Twitter and Reddit. The 39,565 unique comments were labeled by 7912 individual annotators, resulting in a total of 135,556 labeled entries (Berkeley, 2024). The dataset includes over 130 variables, which can be logically grouped into the following feature categories:

Variable Category	Description
Identifiers	Unique identifiers like comment_id and annotator_id to distinguish and reference each entry and annotator
Types of Hate Speech	labels for hate-related dimensions such as humiliate, violence, dehumanize, and genocide, capturing different intensities of hate
Target Groups	Binary indicators for identity-based targeting (e.g. target_race_black, target_gender_women, target_religion_muslim)
IRT-based FeatureS	Variables like annotator_severity and outfits, used to model annotator bias via Item Response Theory
Annotator Background	Demographic information about annotators, including gender, income, education, political ideology, and religion
Platform Information	The platform where the comment was originally posted (e.g. YouTube or Reddit)
Hate Speech Score	A continuous label representing the overall severity of the hatefulness in each comment
Text	The original comment text that serves as the basis for input in NLP models.

The variable hate_speech_score is a continous variable that represents the severity of hate speech content in comments. While analyzing the data, it was found that the maximum value assigned to a comment was 6.3 while the comment with least hate speech was given a score of –8.34. The way this score is interpreted, as explained by Kennedy, higher scores indicate higher

hate spech content, while lower scores represent lower hate speech in comments. (Kennedy et al., 2020) Furthermore, Kennedy also explains that scores above 0.5 are approximately hate speech, between 0.5 and –1 it is likely to be neutral or ambiguous and everything below –1 is likely counter or supportive speech (Kennedy et al., 2020).

Despite the richness of the dataset, we retained only three variables for our analysis: comment_id, text, and hate_speech_score. All other variables were excluded either because they were not relevant for natural language processing tasks or because they contained sensitive demographic information that could raise privacy concerns. By focusing only on the comment content, its unique identifier and its corresponding score, we ensure a clean, ethical, and task-appropriate dataset for training our model on hate speech detection.

Dataset 2: Bluesky Social Media API Stream

The second set of data was constructed using real-time user generated posts, streamed into the Spark environment from the social media platform BlueSky via the python interface to the AT-Protocol, which is what the decentralized platform uses. This "Authenticated Transfer Protocol" can be seen as a pipeline, which we cut into and access the posts directly from a specific domain, instead of the centralized servers (Atproto, 2025). In our project we access the Bluesky feed "hot-classic" as it is one of the most popular ones. The data collected from the BlueSky feed consists of the following ten variables:

Variable Category	Description
Identifiers	uri, cid, author_did and author_handle identify each post and its author uniquely within the decentralized Bluesky environment
Post Content	The content variable contains the actual text of each post and is the central input for NLP tasks
Metadata	created_at_iso and created_at_dt store the post's creation time in string and timestamp formats, reply_count, repost_count, and like_count capture user engagement metrics.

For our analysis, we retained only the variables uri and content. The uri ensures that each post can be uniquely referenced, while the content is the basis for Natural Language Processing and model predictions. The remaining variables were excluded as they either do not contribute to the model training or could contain sensitive user-related metadata. This decision helps reduce the data footprint and protect user privacy. We use this dataset to evaluate the trained models on real-world data and assess their ability to generalize beyond the training corpus

Project Architecture

The pipeline developed in this project is designed to allow data from two different sources: the labeled Dataset retrieved from HuggingFace and the live streamed data from the Social Network BlueSky. Both data sources are processed inside the Spark environment using a modular pipeline composed of filtering, cleaning, and NLP-processing. After pre-processing, the HuggingFace data is split into subsets of training and test data. These subsets are then fed into a machine learning module, where predictive models are created and evaluated. After analyzing the performance factors of each model, the most accurate model based on their respective R^2 and RMSE is selected to be used on the pre-processed data from the BlueSky API stream.

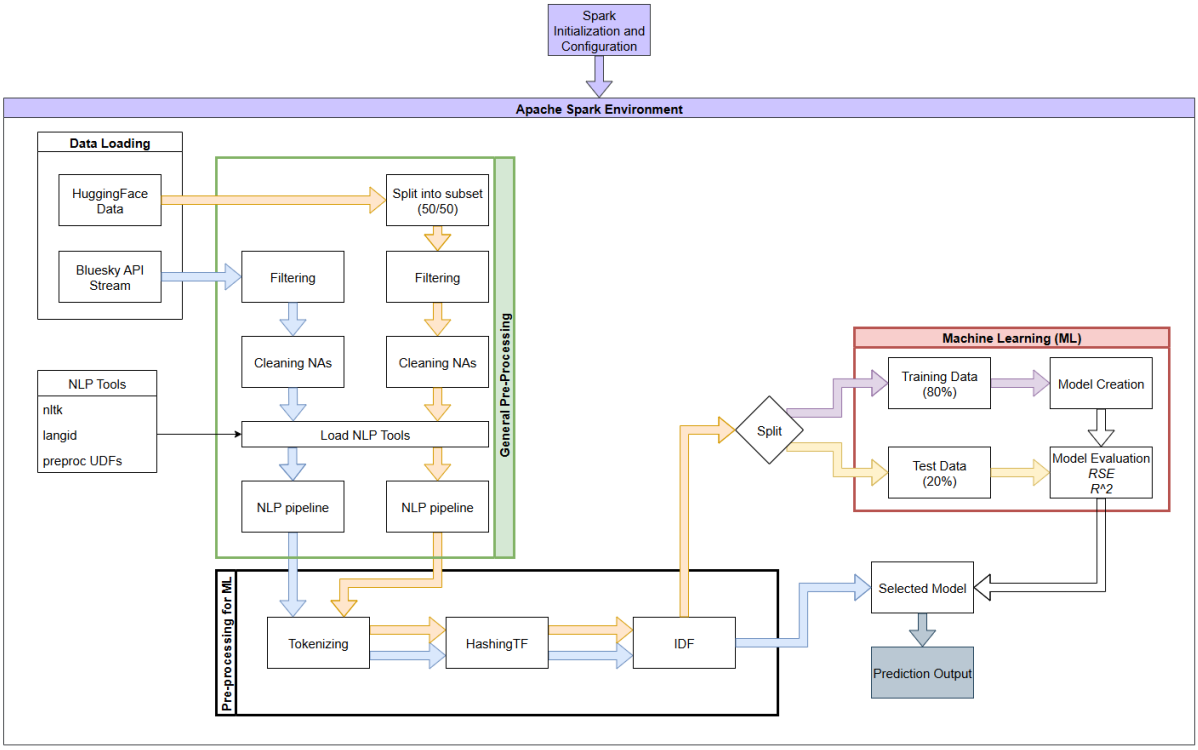


Figure 1: Visual Representation of project architecture

Analysis steps and results

In this section we will take a deeper look at the exact steps that went into preparing the data and creating the models, as well as giving a final evaluation on the models created.

Data Preparation / Pre-processing

Both data sources undergo almost identical preparation processes, except for the initial filtering. Since the datasets come from different sources, each dataset contains different variables. Starting with the data fetched from HuggingFace, the dataset contained over 131 different variables, out of which only three were selected (Berkeley, 2024). To reduce the computational runtime during development, we initially worked with a random subset of just 0.1% of the original dataset. Later, we increased this to 50% to receive meaningful output. This allowed us to maintain enough data points while simultaneously keeping the processing times manageable.

In the case of the Bluesky comments, filtering the columns was rather simple, since we only needed the columns containing text comments and the uri to use as a unique comment identifier. All other variables were discarded, since the accompanying metadata was not relevant for the model prediction and could pose legal and ethical concerns regarding data privacy.

After this filtering process of relevant columns, both datasets undergo the exact same pre-processing. First, the data gets cleaned from any missing values before entering the NLP pipeline with user-defined functions. This pipeline consists of various steps: checking language and filtering only comments in English, removing stop words, removing features, tag-and-

remove, lemmatizing the comments, checking blank spaces, and finalizing with the removal of duplicates to maintain data integrity. These steps are crucial for reducing data dimensionality.

Predictive Model Creation

Now that the data is cleaned and ready to be used, it is time to start creating our predictive models. However, the data still needs to be prepared so that it is in the right format for machine learning. Since our data is in words, we need to transform this into something computer-legible; thus, we applied the tokenization methods discussed in class – namely RegexTokenizer, HashingTF and IDF. The former splits the text into tokens based on expression patterns, then HashingTF converts these into numeric token counts which are then scaled by the latter program. We added all of these to a pipeline and applied it to the training and test data.

As for the models, we decided on ones that can handle complex underlying relationships, since we are dealing with natural language. After consulting the Apache Spark's documentation, and knowledge of machine learning from previous courses, we agreed on using a Random Forest Regression model – a model that creates several decision trees based on the data and averages the results (Apache Spark, 2025a). Similarly, we employed Gradient Boosted Tree Regression, as it works similarly to the Random Forest model but with a loss function it tries to minimize recursively. To have a baseline, we also wanted to create a less sophisticated model, and we thought a linear regression model would suffice (Apache Spark, 2025b).

Testing and Evaluation Methods

Next, we predicted values for the hate speech score of the text in the test data, training our models on the train data. To evaluate which model worked best, we calculated the root mean square error, as suggested in the documentation provided (Spark, 2025c). We also checked the R squared values for model fit and visualized what the models were actually predicting by printing out the top 5 posts with the highest predicted hate speech score.

Results

Here are the values of RMSE and and R^2 from the 3 models:

	Linear Regression	Gradient Boosted Trees	Random Forest
RMSE	1.69	1.82	1.89
R^2	0.33	0.22	0.16

According to both RMSE and R^2, the simple linear model seemed to perform the best. Even so, the RMSE for the linear model is almost 15% of the entire range. A similar issue appears in the R^2 values, as all of them are rather small, meaning the model is not doing much better than just simply predicting the mean. Both of these signify underfitting, which is likely due to a complex underlying relationship in the data. Moreover, it could be that the relationship between hate speech score and text is not that strong, and other factors, such as user id, may play a role. Nonetheless, we shall move forward with the linear model, as it performed the best. Then, we used our best model, the linear one, to predict the hate speech scores of the posts retrieved from Bluesky. To preserve the original posts through the pre-processing, we created row ids so that the original text content of the posts can be printed along with the predicted hate speech scores.

```
Top 5 predictions:
+-------------------------+---------------------+
|content                  |prediction           |
+-------------------------+---------------------+
|emergency alert texas hamstrung cost money fact everyone texas force turn emergency alert phone nobody take flood many kill|-0.004361678772574984|
|people think kid hey kid look dinosaur cool kid wow kid hey kid look dinosaur cool kid plesiosaur cool pliosaurs exist know difference plesiosaur pliosaur let talk pal|-0.5297460167163429  |
|fox news ask trump deport democratic mayoral candidate straight fascist clownery www dailykos com story|-0.5525241663989536  |
|people situation job recovery tool philosophy shame radical acceptance mean stay place belong mean gotta start get|-0.5532839060289276  |
|authority need plan retain rainwater fall say hetherington upland area sink ground river replenish reservoir water table www com content|-0.5653730120061589  |
+-------------------------+---------------------+
only showing top 5 rows
```

Figure 2: Final ouput of Bluesky Predictions

Looking at the final outputs, despite low R^2 values, the predicted values seem to match up to the content of the posts, for example the, the last post about rainwater needing to be retained, has a hate speech score of ~ –0.56, meaning it is neutral speech, which makes sense. Overall, it seems there was not really any hate speech in the Bluesky posts, according to the predictions, since the highest score is ~0, which still falls into the category of neutral or ambiguous.

Legal and Ethical Issues

Licenses

HuggingFace:

The dataset retrieved from HuggingFace contained a cc-by 4.0 license, which allows us to use this dataset for commercial purposes, modify its contents, distribute it and use it privately, so far as copyright guidelines are followed and all performed changes are disclosed. Under this license, the users of its contents, us, need to: accredit the creator(s) by name and other attribution parties, provide a license notice, copyright notice, a disclaimer notice and provide a link to the material (Creative Commons, n.d.). As described in our project proposal, we have created a text file called LICENSE.txt, which contains a copy of the license agreement. Even though the license allows individuals to use it for commercial purposes, there still are restrictions that restrict us from patenting and trademarking the contents of the dataset.

BlueSky data

The API, which allows us to stream posts from the BlueSky platform, contains an MIT license, which has a great number of liberties compared to other licenses. With this MIT license, we are allowed to use, copy, modify, merge, publish, distribute, sublicense or sell the data collected from it (Bluesky Social PBC, 2025). Yet since the license for this API allows us to use their user generated data freely, the BlueSky team takes no liability in the contents of the collected data. We cannot hold the authors of Bluesky liable for any claim, damages or other liability, whether in an action of contract, tort or otherwise, arising from or in connection to the software or the user or other dealings in the software. (Bluesky Social PBC, 2025) The posts and profiles of Bluesky users are public, and we therefore have access to them. The Bluesky platform states compliance with the GDPR (Bluesky Social PBC, 2024).

GDPR Compliance and Safeguards:

HuggingFace

The utilization of the HuggingFace dataset, which contains labeled hat speech comments sourced across YouTube, Twitter and Reddit, is grounded on a lawful basis under the General Data Protection Regulation (GDPR). In Art. 6.1, processing is only permitted, as long as one or more justifications apply to the instance (European Parliament & Council of the Europe Union, 2016). In our case, our legal justification falls under Art 6.1(f) which allows us to process data on the basis of legitimate interest. On this context, our legitimate interest lies in doing scientific research aimed at creating tools capable of detecting harmful online content. This research topic is of great academic and societal value, since it covers topics related to digital safety, platform moderation, and mental health (Dreißigacker et al., 2024). Due to the fact that the dataset may contain sensitive information related to protected characteristics - like race, religion and gender – our project engages with Art. 9.2(j) GDPR, which allows us to use special category data when necessary for scientific research purposes, as long as the necessary safeguards are implemented (European Parliament & Council of the Europe Union, 2016).

BlueSky

All retrieved data originates from public posts made by users, and the platform itself adheres to the GPDR principles (Bluesky Social PBC, 2024). In line with data minimization principles, direct personal identifiers like usernames and account handles are not kept in our dataset. Furthermore, we ensure that any indirect personal identifiers, like "@" mentions, are systematically removed during preprocessing. Since data containing potentially identifiable information is deleted before model processing, techniques that serve to preserve anonymity like K-anonymity or KTL are not applied.

Safeguards and Data Minimization Measures

To ensure compliance with the principle of data minimization discussed in Art. 5.1(c) GPDR, we implemented multiple safeguards across both datasets utilized throughout this project. The datasets were accessed exclusively on a controlled Jupyter environment, and raw data was not shared or distributed to external parties. Access to the raw data was limited to the research team members, although no formal access control measures were put into practice. For the HuggingFace dataset we deleted all personal information related to annotators that could enable indirect reidentification. For the Bluesky dataset, we remove all "@" mentions and usernames from posts, thereby reducing the risk of identifying user accounts. These steps were taken to reduce the probability of exposing sensitive and personal data and are in line with the best practices for data protection under GDPR regulation.

Transparency and Accountability

We recognize the importance of transparency when dealing with sensitive data, such as the user created comments we parsed from Bluesky. To ensure transparency for any and all stakeholders, we documented all our code in markdown in a clear and consistent manner, so any processing or analysis done can be traced. With any big data project, it is important to keep a high level of accountability – which we established by closely collaborating on our project, meaning throughout its life cycle, it was handled ethically, bolstered by the previously mentioned transparency that ensures traceability.

Bias and Fairness

When it comes to a big data project, bias and fairness are always a concern. One such issue with our project is selection bias, as we are training our models on a specific dataset that might be very narrow and unrepresentative, in other words, might not be applicable to the whole population of social media posts. Moreover, hate speech is in and of itself a subjective concept, as one post might be classified as offensive to one person in one socio-cultural context and not to someone in another. By filtering for only English results, and with natural language processing tools, we reduce this cultural bias by selecting a specific context to apply our model to. In this way, we aspire to be fair and unbiased with our data processing.

Challenges and Experience Gained

Model Selection and Regression Setup

Borisz oversaw the pre-processing associated with machine learning, the creation of ML models themselves and predictions plus display of results. The main issues encountered boiled down to selecting the right models, since the project could have also been a categorization issue, but due to the nature of the train file, he decided to use regression models. To understand what regression models would be ideal, he studied the Spark ML documentation. Another major issue encountered was that the code took a long time to run, and with a reduced sample size, models would not be able to function as they should, due to the NLP nature of the task. His key takeaways were new familiarity with regression models in the Spark environment and machine learning for big data in general, and what metrics can be used to evaluate model fit.

Loading of Bluesky data

The code to access the Bluesky posts was first written by Matthäus in Google Collab, and later transferred to Jupyter. It worked without issues for him, but others encountered errors with the library, or rather the individual parts of the Atproto library and the python version used in Jupyter. It seems like some parts of Atproto were stored in the memory of the other team members, so installing the package normally wouldn't overwrite that version. Our two options were to create a virtual environment, but we saw this as too complex and unnecessary for this assignment, so we decided to rewrite the code that loads Atproto (and some other packages). The new version purges Atproto fully from the disk and memory and then reinstalls a specific version of it. This version now works for everyone. In future shared projects it would be useful to create a helper function, with which you can define the exact package version that is needed, which then purges any potential previous mentions from the system. The key learning therefore was that you need to make your code more robust and account for these issues, especially if you want to open-source it and make it available for others.

Speed of the Code

Maximilian was responsible for the Spark setup, the loading of the HuggingFace dataset, and the entire preprocessing including both NLP pipelines. Since the dataset from HuggingFace was very large, he encountered several issues with the speed of the code. Initially, only 0.1% of the data was loaded, but later 50% was needed, which took more than 10 hours to process. At that point, Maximilian decided to test two approaches in parallel. On one side, he set up a private Jupyter server, while on the other, he created a local environment and ran the code on

his own machine. Both methods were significantly faster, with runtimes of about one hour. The Jupyter server was later abandoned, as it was still slower and caused errors due to conflicting Python versions installed on the machine. However, the code continued running, so the issue turned out to be only visual. The key takeaway is that NLP processes can be time-consuming, and it is essential to consider this – especially when working with large datasets and under strict deadlines.

Legal and ethical research

Martin was responsible for researching and applying legal and ethical aspects for the project, with a strong focus on GDPR compliance, data minimization, and responsible use of user-generated data. While no major issues were encountered during the research process, it rapidly became clear that this section demanded close attention to detail, since misinterpreting sections of legal frameworks could lead to serious consequences. This part of the project greatly undermined the importance of combining technological solutions with ethical and legal standards, as to ensure responsible and safe data processing in machine learning applications.

Recommendations for Future Work

This project has many potential uses, and we consider its further development to be not just lucrative, but an important endeavor. By creating an open-source hate speech detector, we empower smaller platforms and independent developers to implement this functionality with little adaptation. For example, Bluesky feed moderators could easily use the algorithm or customize it with their own lists of unwanted phrases. There are key technical requirements: anyone adapting or hosting the algorithm must have sufficient computational capacity. While the load per post is small, it can add up quickly on a social media platform, and moderators needing real-time detection would need to stream data to the algorithm continuously.

Declaration of Generative AI

During the preparation of this work the author(s) used OpenAI's ChatGPT 4o in order to structure contents of report, do faster research on specific topics, and summarize documents for faster reading comprehension. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication. It was also used to help structure and write the README.txt file.

Furthermore, the authors used "Languagetool", "Grammarly" and the built-in spellcheck of Microsoft word to correct the text of this report.

They also used LLM's such as Gemini 2.5 pro as a last resort option to clear up errors, for which they were not able to find online resources.

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